
COMPUTER SYSTEMS ADMINISTRATION

1. Introduction

Computer systems administration is the practice of managing, maintaining, and securing computer systems and networks so they run smoothly and reliably. It is usually carried out by a system administrator (sysadmin).

Tasks of a System Administrator

i) Setting up systems

- Installing operating systems (Windows, Linux, etc.)
- Configuring servers, computers, and network devices

ii) Maintaining systems

- Applying updates and patches
- Monitoring performance (speed, storage, uptime)
- Fixing technical problems when they occur

iii) Managing users

- Creating and deleting user accounts
- Setting permissions (who can access what)
- Ensuring proper authentication and security

iv) Network management

- Configuring routers, switches, and connections
- Ensuring systems can communicate properly
- Managing IP addresses and connectivity

v) Security management

- Protecting systems from viruses and cyberattacks
- Setting up firewalls and access controls
- Backing up data and planning recovery in case of failure

vi) Automation and scripting

- Writing scripts to automate repetitive tasks (e.g., backups or updates)

2. Servers

2.1 Server

A server is a computer or system that provides services, data, or resources to other computers (called clients) over a network.

Types of servers

- Web server → delivers websites
- Application server → runs application logic
- Database server → stores and retrieves data
- File server → stores files
- Email server → sends/receives emails

2.2 Server Architecture

Server architecture refers to the design and structure of how servers are set up, organized, and interact with clients and other systems to deliver services. It defines how requests are handled, where data is stored, and how components communicate.

Key Components of Server Architecture

- Clients: Front-end user devices (laptops, phones) that initiate requests.
- Server/Service Provider: The backend system (physical, virtual, or cloud) that processes requests and stores data.
- Network: The communication channel (Internet, intranet) connecting clients and servers.
- Protocols: Standardized rules like HTTP/HTTPS or SQL for client-server communication.

Common Models of Server Architecture

- Client-Server Pattern: Partitions workloads between service providers (servers) and service requesters (clients).
- Peer-to-Peer (P2P) Architecture: Direct connections between devices, each acting as both client and server. Simple setup but lacks centralized management.
- Distributed Architecture: Multiple geographically dispersed servers working collaboratively, providing fault tolerance and improved performance.
- Microservices Architecture: Applications split into small, independent services communicating via APIs. Enhances scalability and fault isolation.
- N-Tier Pattern: Variation of client-server where presentation, application processing, and data services are separated into distinct tiers.

2.3 Key Considerations in Designing Server Architecture

Factor	Description
Load Balancing	Distributes network traffic across multiple servers, preventing overload and ensuring consistent performance.

Factor	Description
Scalability	Handles increasing demands via vertical scaling (upgrading hardware) or horizontal scaling (adding servers).
Fault Tolerance	Minimizes downtime through redundancy, failover mechanisms, and backup systems.
Cost Efficiency	Uses virtualization, cloud services, and resource optimization to minimize costs.
Monitoring & Management	Implements monitoring tools to track metrics, identify issues, and automate management tasks.

2.4 Types of Server Hardware

- Tower servers: Standalone upright case; easy cooling, inexpensive, favored by small businesses. Takes up more physical space.
- Rack servers: Mounted on server racks in data centers; takes less space than tower servers; built to consistent size standards.
- Blade servers: Modular architecture – multiple thin blades in a shared chassis; chassis provides power, cooling, and networking.

2.5 Server Installation & Deployment

i) Planning

- Identify the purpose of the server (hosting a website, database, or application).
- Estimate the number of users and expected traffic.
- Decide performance requirements (CPU, RAM, storage).
- Choose hosting: Cloud platforms (AWS) or physical/on-premise servers.
- Plan for scalability, security, and budget.

ii) Server Setup (Provisioning)

- Select an operating system (e.g., Ubuntu Server).
- Create the server instance and assign IP address, storage, memory, and CPU.
- Configure remote access (usually SSH).

iii) Software Installation

- Web server: Nginx or Apache
- Database: MySQL or PostgreSQL
- Runtime environment: Node.js, Python, or Java

iv) Security & Network Configuration

- Configure firewall (allow only necessary ports: 22, 80, 443).
- Disable root login and use secure authentication.
- Set up a domain and DNS.
- Enable HTTPS using Let's Encrypt.

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- Install security tools (e.g., intrusion prevention).

v) Application Deployment

- Upload application files (via Git, FTP, or SCP).
- Install dependencies and configure environment variables.
- Use automation tools like GitHub Actions for consistent deployment.

vi) Maintenance & Monitoring

- Monitor performance (CPU, memory, uptime) using tools like Prometheus.
- Set up logging to track errors and activity.
- Perform regular updates and patches.
- Backup data regularly and test recovery procedures.

3. Install and Configure a Virtual Machine

A Virtual Machine (VM) is software that creates a simulated computer inside your real system, allowing multiple operating systems to run independently on the same physical hardware. It virtualizes resources like CPU, RAM, storage, and network.

- Allows running multiple operating systems on a single physical machine.
- Each VM works as an independent computer with its own OS and resources.
- Common VM tools include VirtualBox, VMware, and Hyper-V.

3.1 Downloading and Installing VirtualBox

Follow these steps to download and install VirtualBox:

1. Go to the official VirtualBox website and download the Windows Hosts installer (.exe file).
2. Locate the downloaded .exe file and double-click to launch the Setup Wizard.
3. Click Next on the welcome screen to begin installation.
4. Choose installation location (or keep default) and click Next.
5. Select shortcuts and file associations, then click Next.
6. Click Install to begin copying files.
7. If prompted to install device software/certificates, click Install or Allow.
8. Click Finish. Optionally check 'Start Oracle VM VirtualBox after installation'.

3.2 How to Install Ubuntu on VirtualBox

9. Open VirtualBox and click New.
10. Choose guest OS type (Linux) and version (Ubuntu), then click Next.
11. Set Base Memory (RAM) and click Next.
12. Set VM storage size (recommended: 30 GB or more) and click Create.
13. Select your VM and click Start. A 'First Run Wizard' will appear.
14. Click the folder icon and select your Ubuntu .iso file, then click Next.
15. Click Start to boot from the ISO and follow the Ubuntu installation wizard.
16. After installation, remove the .iso from the virtual optical drive before rebooting.

4. Introduction to the Linux Operating System

Linux is a Unix-like open-source operating system developed in 1991 by Linus Torvalds. It was originally created for Intel x86 personal computers and has since been ported to more platforms than any other OS. More than 90% of the world's 500 fastest supercomputers run Linux. The Android system, widely used on mobile devices, is built on the Linux kernel.

4.1 Basic Features

Feature	Description
Portable	Supports installation on any hardware platform.
Open Source	Source code freely available; community-based development.
Multituser	Multiple users can access system resources simultaneously.
Multiprogramming	Multiple applications can run simultaneously.
Hierarchical File System	Standard file structure for organizing system and user files.
Shell	Interpreter program for executing OS commands.
Security	Password protection, file access controls, and data encryption.

4.2 Linux Advantages

- Low cost: Free under GNU General Public License; large free software repositories.
- Stability: Does not need periodic reboots; uptime of hundreds of days is common.
- Performance: High performance on workstations and networks; handles many simultaneous users.
- Network friendliness: Strong built-in networking support for both client and server roles.
- Flexibility: Usable for servers, desktops, and embedded systems.
- Compatibility: Runs common UNIX software and processes common file formats.
- Security: Secure file permissions, firewall support, and safe software repositories.
- Open Source: Full source code access for developers.

4.3 Linux Drawbacks

- Hardware drivers: Fewer drivers for peripheral hardware compared to Windows.
- Software alternatives: Popular tools like Photoshop and MS Office are not natively available.
- Learning curve: Command-line interface can be complex for beginners.
- Games: Most games are developed for Windows, not Linux.

4.4 Linux System Architecture

Layer	Description
Hardware	Physical components: CPU, monitor, keyboard, hard disk, and other connected devices.
Kernel	Core of the OS; directly interacts with hardware. Manages memory, schedules processes, and provides system calls. Loaded at boot (/boot). Written in C.
Shell	Interface between user and kernel. Interprets and executes user commands. Use <code>echo \$SHELL</code> to see current shell.
Applications	User-facing software (terminal, browser, LibreOffice). Communicates with kernel via system calls.
System Utilities	Programs for managing processes, controlling user accounts, and configuring system settings.

4.5 Linux File System Hierarchy (FHS)

Linux uses a hierarchical file system based on the Filesystem Hierarchy Standard (FHS). Every file and directory is located under the root directory `/`.

Directory	Purpose
<code>/</code>	Root of the entire filesystem. Starting point for all paths.
<code>/bin</code>	Essential user binaries (<code>ls</code> , <code>cp</code> , <code>mv</code> , <code>cat</code>). Available in single-user mode.
<code>/sbin</code>	System binaries requiring admin privileges (<code>fsck</code> , <code>reboot</code> , <code>ifconfig</code>).
<code>/lib</code>	Shared libraries needed by <code>/bin</code> and <code>/sbin</code> .
<code>/home</code>	User home directories (e.g., <code>/home/alice</code>).
<code>/root</code>	Home directory of the root (admin) user.
<code>/etc</code>	System-wide configuration files (<code>/etc/passwd</code> , <code>/etc/shadow</code> , network configs).
<code>/var</code>	Variable data that changes frequently: logs, email queues, web server files.
<code>/tmp</code>	Temporary files; typically cleared on reboot.
<code>/dev</code>	Device files representing hardware (<code>/dev/sda</code> , <code>/dev/tty</code> , <code>/dev/null</code>).
<code>/proc</code>	Virtual filesystem with kernel/process information (<code>/proc/cpuinfo</code> , <code>/proc/meminfo</code>).
<code>/sys</code>	Kernel and hardware interface for tuning and device configuration.
<code>/usr</code>	User applications and libraries (<code>/usr/bin</code> , <code>/usr/lib</code> , <code>/usr/share</code>).
<code>/boot</code>	Boot loader files (kernel, GRUB). Required for system startup.
<code>/mnt</code>	Temporary manual mount points.
<code>/media</code>	Automatically mounted removable media (USB, CDs).
<code>/opt</code>	Optional/third-party software (e.g., Google Chrome, Oracle software).

4.6 File Colors in Linux Terminal

- Blue – Directory
- Green – Executable or recognized data file
- Cyan (Sky Blue) – Symbolic link file
- Yellow with black background – Device file
- Magenta (Pink) – Graphic image file
- Red – Archive file
- Red with black background – Broken link
- White – Regular file

4.7 Ubuntu Linux Commands

Commands follow this general format:

\$ **Command** **[options]****[arguments]**

Navigation Commands

Command	Description	Example
ls	List files and folders in the current directory.	ls -la
pwd	Show the path of the current working directory.	pwd
dir	Print all directories in the current directory.	dir
cd	Change directory.	cd Desktop

Is Command Options

Option	Description
-a or --all	Lists all files including hidden ones.
-l	Display results in a detailed/long format.
-h or --human-readable	Display file sizes in human-readable format (KB, MB).
-S	Sort results by file size.
-t	Sort results by modification date.
-r or --reverse	Sort results in reverse order.
-R	Print subdirectories recursively.

File Management Commands

Command	Description	Example
touch	Create a new file or update file timestamp.	touch file.txt
nano	Open text editor for editing files. Close with CTRL+X.	nano file.txt

Command	Description	Example
cat	Display file content or combine files.	cat file.txt
mkdir	Create a new directory.	mkdir BIT
rm	Remove a specific file.	rm output.txt
rmdir	Remove an empty directory.	rmdir BIT
rm -r	Remove directory and all its contents recursively.	rm -r medy
cp	Copy files or directories.	cp file.txt Desktop/
mv	Move or rename files/directories.	mv medy Desktop/
wget	Download content from the internet.	wget https://example.com/file
history	Show list of previously executed commands.	history
grep	Search for a pattern in files.	grep -i "hello" notes.txt

5. Package Management with APT and dpkg

On Debian-based Linux distributions (Ubuntu, Debian), package management is handled primarily by APT (Advanced Package Tool) and dpkg (Debian Package).

5.1 APT – Advanced Package Tool

APT simplifies software management by installing, updating, and removing packages and their dependencies automatically.

Command	Description
sudo apt update	Update the local package index.
sudo apt upgrade	Upgrade all installed packages.
sudo apt install package_name	Install a package from online repositories.
sudo apt remove package_name	Remove an installed package.
sudo apt install ./package.deb	Install a local .deb file (from current directory).
sudo apt install /complete/path/package.deb	Install a local .deb file using full path.

5.2 dpkg – Debian Package Manager

dpkg is the low-level package manager. It installs and manages .deb packages directly but does not automatically resolve dependencies.

Command	Description
sudo dpkg -i package.deb	Install a .deb package.

Command	Description
<code>sudo dpkg -r package_name</code>	Remove a package (keep config files).
<code>sudo dpkg -P package_name</code>	Remove package and its configuration files.
<code>dpkg -l</code>	List all installed packages.
<code>dpkg -s package_name</code>	Check if a package is installed.
<code>dpkg -L package_name</code>	Show files installed by a package.
<code>dpkg -S /path/to/file</code>	Find which package owns a file.
<code>dpkg -I package.deb</code>	Display package information.
<code>sudo apt --fix-broken install</code>	Fix broken/missing dependencies.

Common package states shown by `dpkg -l`:

- `ii` – Installed successfully
- `rc` – Removed but configuration files remain
- `un` – Package not installed

6. Linux Operating System Remote Login (SSH)

Remote login to Linux is achieved using the Secure Shell (SSH) protocol via the OpenSSH package.

Component	Function
OpenSSH Server (<code>sshd</code>)	Allows other computers to connect to your machine.
OpenSSH Client (<code>ssh</code>)	Allows your machine to connect to other systems.
PuTTY	Windows SSH client used to connect to Linux servers remotely.

6.1 Installing and Configuring OpenSSH Server

17. Install the SSH server:

```
sudo apt update
sudo apt install openssh-server
```

18. Check installed version:

```
sshd -v
```

19. Start and enable the service:

```
sudo systemctl start ssh
sudo systemctl enable ssh
```

20. Check service status:

```
sudo systemctl status sshd
```

6.2 Connecting via PuTTY (from Windows)

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21. Open PuTTY on the host computer.
 22. Enter the server hostname or IP address (e.g., 192.168.56.3).
 23. Select SSH as connection type and click Open.
 24. Enter the server username and password when prompted.

7. Configuring a File Server with Samba

Samba enables an Ubuntu system to share files and printers with Windows, Linux, and macOS computers via the SMB/CIFS protocol.

7.1 Installing and Configuring Samba

25. Update packages and install Samba:

```
sudo apt update  
sudo apt install samba
```
26. Verify installation:

```
samba --version
```
27. Create a shared directory:

```
sudo mkdir -p /srv/samba/shared
```
28. Set permissions:

```
sudo chmod 2777 /srv/samba/shared  
sudo chown nobody:nogroup /srv/samba/shared
```
29. Back up the Samba configuration file:

```
sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.bak
```
30. Edit the Samba configuration:

```
sudo nano /etc/samba/smb.conf
```

Add the following section at the end:

```
[Shared]  
path = /srv/samba/shared  
browseable = yes  
writable = yes  
guest ok = no  
read only = no  
create mask = 0777  
directory mask = 0777
```

Samba Configuration Options

Option	Meaning
path	Directory being shared
browseable	Visible on the network

Option	Meaning
writable	Allows writing
guest ok	Whether anonymous access is allowed
read only	Prevents or allows modifications
create mask	Default permissions for files
directory mask	Default permissions for directories

31. Add a Samba user and set password:
`sudo smbpasswd -a username`
32. Enable the Samba account:
`sudo smbpasswd -e username`
33. Test configuration for syntax errors:
`testparm`
34. Restart and enable Samba services:
`sudo systemctl restart smbd`
`sudo systemctl restart nmbd`
`sudo systemctl enable smbd nmbd`
35. Allow Samba through the firewall (if UFW enabled):
`sudo ufw allow samba`

7.2 Accessing Samba Shares

From Windows

36. Enable network discovery in Windows settings.
37. Open File Explorer and enter the server path:
`\\192.168.56.3\shared`
38. Enter Samba username and password when prompted.

From Linux Desktop

39. Open File Explorer (Files app).
40. Enter the Samba path in the location bar:
`smb://192.168.56.3/shared`
41. Enter credentials when prompted.

8. Lab Reference – Virtual Network Setup

Lab 1: Setting Up a Virtual Server Network

This lab sets up a virtual network with DNS, web, DHCP, file, and database servers hosted in Oracle VirtualBox on Windows.

Computer/Host Name	IP Address	Purpose
database-web-server	192.168.56.4	Database and web server
file-dhcp-server	192.168.56.3	File and DHCP Server
dns-server	192.168.56.2	DNS Server
windowspc	192.168.56.1	Host computer with Oracle VirtualBox installed

Key Configuration Steps

42. Configure two Ethernet adapters per VM: Adapter 1 = NAT (internet), Adapter 2 = Host-Only (LAN).
43. Set static IPs by editing /etc/netplan/99_config.yaml and running: `sudo netplan apply`
44. Set hostnames using: `sudo hostnamectl set-hostname <name>`
45. Test internet connectivity: `ping www.google.com`
46. Test inter-server connectivity: `ping <server-IP>`

TEST FOR PRACTICE

Choose the correct answer by writing its corresponding letter. Each question carries 1 mark.

Q1: Which among the following interacts directly with system hardware?

- A. Shell
- B. Commands
- C. Kernel ✓ CORRECT**
- D. Applications

Explanation:

The Kernel is the core component of the Linux OS that directly interacts with hardware. It manages memory, schedules processes, and handles system calls. The Shell acts as an interface between the user and kernel, while Applications communicate through system calls.

Reference: Section 4.4 (Linux System Architecture) – Class Notes: "The kernel is the core component of the Linux OS and directly interacts with hardware."

Q2: Which option of the ls command is used to view all files and folders, including hidden ones?

- A. -l
- B. -b
- C. -a ✓ CORRECT**
- D. -i

Explanation:

The -a (or --all) option lists ALL files and directories, including hidden ones (files starting with a dot). The -l option shows detailed format, -S sorts by size, and -t sorts by date.

Reference: Section 4.7 (Ubuntu Linux Commands – ls Command Options) – Class Notes: "-a or --all: Lists all files and directories, including hidden ones."

Q3: What is the purpose of the pwd command?

- A. To display the current working file
- B. To display the current working location ✓ CORRECT**
- C. To display the current working folder

D.	To display the current path to the file
----	---

Explanation:
The pwd (present working directory) command shows the full path from the root to the current directory you are working in – i.e., your current working location in the filesystem.
Reference:Section 4.7 (pwd Command) – Class Notes: "This command refers to the current working directory you are operating in. It shows the path from the root to the current working directory."

Q4: How do you get help about the command "cp"?	
A.	help cp
B.	man cp ✓ CORRECT
C.	cd
D.	None of the mentioned

Explanation:
The man (manual) command provides detailed help and documentation for any Linux command. Running 'man cp' opens the manual page for the cp command, showing all its options, syntax, and usage examples.
Reference:General Linux Command Reference – Class Notes Section 4.7: The man command is the standard way to access command documentation in Linux/Ubuntu.

Q5: Who founded Linux Kernel?	
A.	Ken Thompson and Dennis Ritchie
B.	Linus Torvalds ✓ CORRECT
C.	Linus Torvalds and Ken Thompson
D.	Richard Stallman

Explanation:
In 1991, Linus Torvalds, a student at Helsinki University, created the Linux Kernel as a freely accessible version of Unix. He released it on October 5, 1991. Ken Thompson and Dennis Ritchie created Unix in 1969, while Richard Stallman founded the GNU Project.
Reference:Section 4 (Introduction to the Linux OS) – Class Notes: "In 1991, Linus Torvalds (a student at Helsinki University) decided to work on a freely accessible version of Unix known as Linux... he defined a Kernel as a component of Linux and released it on October 5, 1991."

Q6: Which symbol represents the user's home directory?

A. ~ ✓ CORRECT

B. @

C. #

D. /

Explanation:

In Linux, the tilde symbol (~) represents the current user's home directory. For example, ~ is equivalent to /home/username. The / symbol represents the root directory, while # indicates the root user prompt, and @ is used in email addresses and other contexts.

Reference: Section 4.5 (Linux File System Hierarchy) – Class Notes: "/home – Each user gets a personal folder (e.g., /home/alice)." The ~ shortcut navigates to this personal home directory.

QUESTION TWO – Linux Directory Structure

Study Figure 1 (Linux directory structure) and answer the questions below.

Q2(i) (02 Marks): What is the function/purpose of the /var directory?

Answer:

The /var directory stores variable data – that is, files and data that change frequently during normal system operation.

Its key functions include:

- Storing system log files (e.g., /var/log/syslog, Apache logs, MySQL logs, SSH logs, boot logs)
- Holding email queue data
- Storing web server files
- Keeping cache data and spool files
- Managing print queues and other service data

The system must be able to write to /var during operation, making it one of the most actively used directories. Logs stored in /var/log are especially important for system debugging and monitoring.

Reference: Section 4.5 (Linux File System Hierarchy – /var) – Class Notes: "/var – Variable files: Stores data that changes frequently. All log files are available including system logs, Apache, MySQL, remote connection logs, SSH, and boot logs."

Q2(ii) (02 Marks): Write a Linux command to copy a file named BIT allocated in directory /usr/bin/ to the directory /var/log/

Answer:

The cp (copy) command is used to copy files from one location to another.

Command syntax:

cp [source] [destination]

Answer:

Reference: Section 4.7 (cp Command) – Class Notes: "The cp command will help you to copy any file or folder to any directory."

cp /usr/bin/BIT /var/log/

Breakdown:

Part	Meaning
cp	The copy command
/usr/bin/BIT	Source – full path to the file BIT in /usr/bin/
/var/log/	Destination – the /var/log/ directory

Q2(iii) (02.5 Marks): Imagine you are in the root directory (/). Write a Linux command to navigate to the directory /usr/local/

Answer:

The cd (change directory) command is used to navigate between directories.

Since we are already at the root directory (/), we can use an absolute path:

Answer:

Reference: Section 4.7 (cd Command) – Class Notes: "You can change the directories in the terminal using the 'cd' command... You can navigate to any directory from the present working directory by specifying the path by imitating the root directory."

cd /usr/local/

Note: Since we are in the root directory (/), both of these are equivalent and correct:

cd /usr/local/

`cd usr/local/`

The absolute path `cd /usr/local/` is preferred as it works from any directory, not just root.

Summary of Answers

Q#	Question (short)	Correct Answer
1	Interacts directly with hardware?	C. Kernel
2	ls option for hidden files?	C. -a
3	Purpose of pwd command?	B. Current working location
4	Get help on cp command?	B. man cp
5	Who founded Linux Kernel?	B. Linus Torvalds
6	Symbol for home directory?	A. ~
2(i)	/var directory purpose?	Stores variable/changing data & logs
2(ii)	Copy BIT from /usr/bin/ to /var/log/	<code>cp /usr/bin/BIT /var/log/</code>
2(iii)	Navigate from / to /usr/local/	<code>cd /usr/local/</code>